AN EXAMPLE ARTICLE*

DIANNE DOE†, PAUL T. FRANK‡, AND JANE E. SMITH‡

Abstract. This is an example SIAM LaTeX article. This can be used as a template for new articles. Abstracts must be able to stand alone and so cannot contain citations to the paper's references, equations, etc. An abstract must consist of a single paragraph and be concise. Because of online formatting, abstracts must appear as plain as possible. Any equations should be inline.

Key words. example, LaTeX

AMS subject classifications. 68Q25, 68R10, 68U05

1. Introduction.

The introduction introduces the context and summarizes the manuscript. It is importantly to clearly state the contributions of this piece of work. The next two paragraphs are text filler, generated by the `lipsum` package.

The paper is organized as follows. Our main results are in section 2, our new algorithm is in section 3, experimental results are in section 4, and the conclusions follow in section 6.

2. Main results.

Here we state our main result as Theorem 2.1; the proof is deferred to **??**.

THEOREM 2.1 (LDL^T Factorization [1]). *If $A \in \mathbb{R}^{n\times n}$ is symmetric and the principal submatrix $A(1:k, 1:k)$ is nonsingular for $k = 1 : n-1$, then there exists a unit lower triangular matrix L and a diagonal matrix*

$$D = \operatorname{diag}(d_1, \ldots, d_n)$$

such that $A = LDL^T$. The factorization is unique.

THEOREM 2.2 (Mean Value Theorem). *Suppose f is a function that is continuous on the closed interval $[a, b]$. and differentiable on the open interval (a, b). Then there exists a number c such that $a < c < b$ and*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

In other words,

$$f(b) - f(a) = f'(c)(b - a).$$

Observe that Theorems 2.1 and 2.2 and Corollary 2.3 correctly mix references to multiple labels.

COROLLARY 2.3. *Let $f(x)$ be continuous and differentiable everywhere. If $f(x)$ has at least two roots, then $f'(x)$ must have at least one root.*

Proof. Let a and b be two distinct roots of f. By Theorem 2.2, there exists a number c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{0 - 0}{b - a} = 0. \qquad \square$$

*Submitted to the editors DATE.

Funding: This work was funded by the Fog Research Institute under contract no. FRI-454.

†Imagination Corp., Chicago, IL (ddoe@imag.com, http://www.imag.com/~ddoe/).

‡Department of Applied Mathematics, Fictional University, Boise, ID (ptfrank@fictional.edu, jesmith@fictional.edu).

This manuscript is for review purposes only.

Note that it may require two LaTeX compilations for the proof marks to show.

Display matrices can be rendered using environments from `amsmath`:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{2.1}$$

Equation (2.1) shows some example matrices.

We calculate the Fréchet derivative of F as follows:

$$\begin{aligned} F'(U,V)(H,K) &= \langle R(U,V), H\Sigma V^T + U\Sigma K^T - P(H\Sigma V^T + U\Sigma K^T)\rangle && \text{(2.2a)} \\ &= \langle R(U,V), H\Sigma V^T + U\Sigma K^T\rangle \\ &= \langle R(U,V)V\Sigma^T, H\rangle + \langle \Sigma^T U^T R(U,V), K^T\rangle. && \text{(2.2b)} \end{aligned}$$

Equation (2.2a) is the first line, and (2.2b) is the last line.

3. Algorithm.

Our analysis leads to the algorithm in Algorithm 3.1.

Algorithm 3.1 Build tree

```
Define P := T := {{1}, ..., {d}}
while #P > 1 do
    Choose C' ∈ C_p(P) with C' := argmin_{C∈C_p(P)} ϱ(C)
    Find an optimal partition tree T_{C'}
    Update P := (P\C') ∪ {⋃_{t∈C'} t}
    Update T := T ∪ {⋃_{t∈τ} t : τ ∈ T_{C'}\L(T_{C'})}
end while
return T
```

4. Experimental results.

Figure 1 shows some example results. Additional results are available in the supplement in **??**.

FIG. 1. *Example figure using external image files.*

5. Discussion of $Z = X \cup Y$.

6. Conclusions.

Some conclusions here.

Appendix A. An example appendix.

LEMMA A.1. *Test Lemma.*

Acknowledgments. We would like to acknowledge the assistance of volunteers in putting together this example manuscript and supplement.

REFERENCES

[1] G. H. GOLUB AND C. F. VAN LOAN, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, 4th ed., 2013.